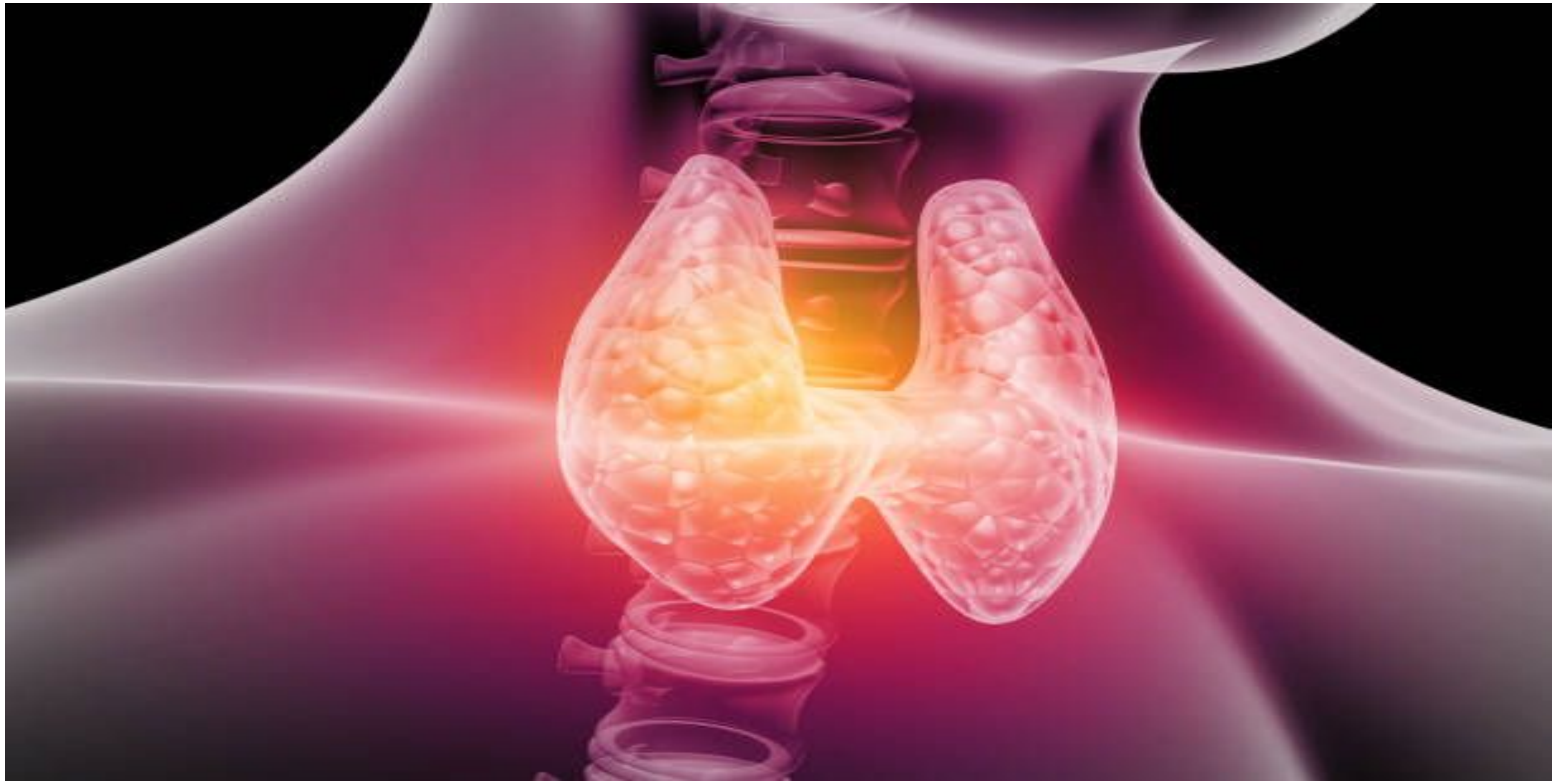
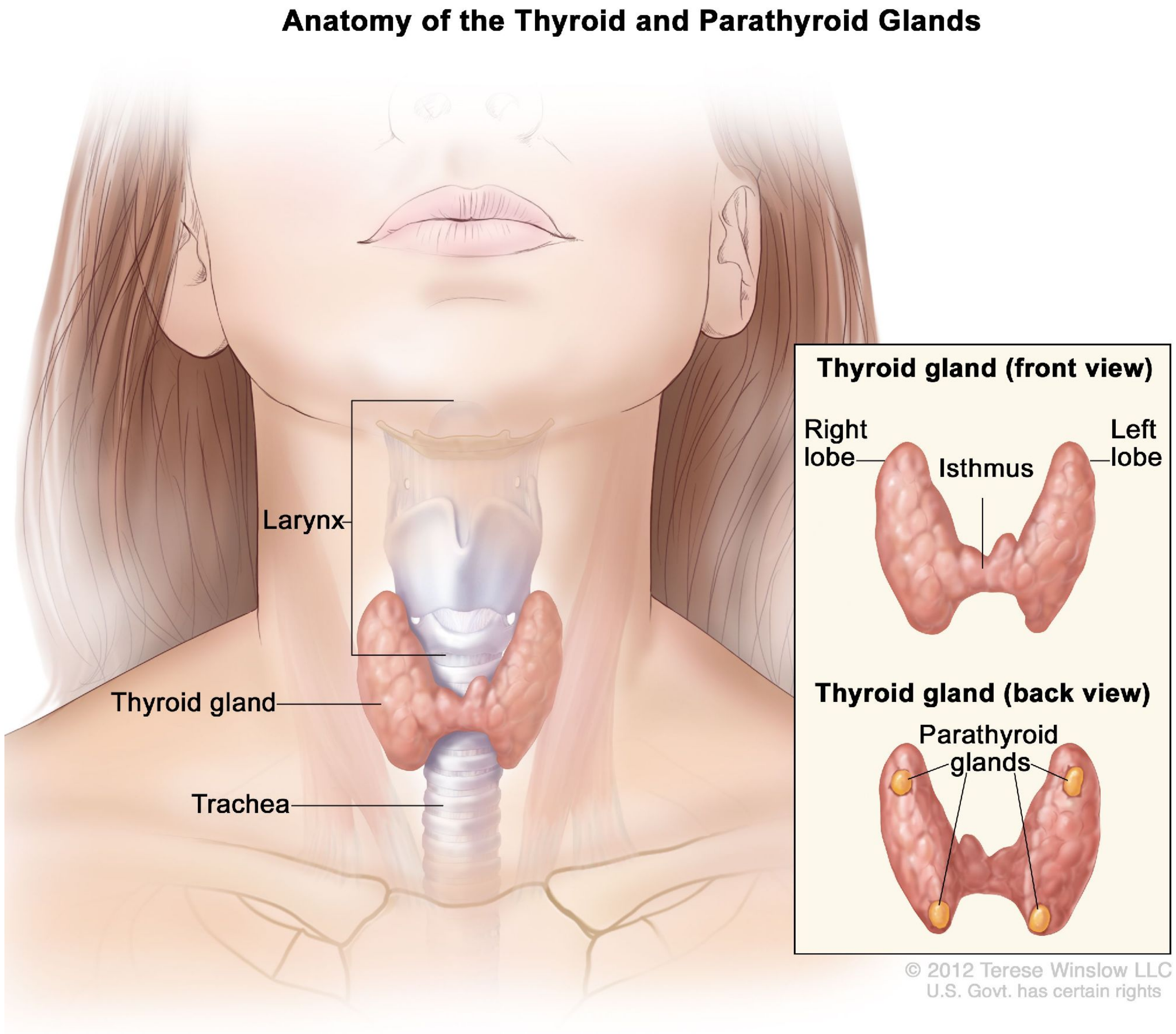
Gross Anatomy & Histology of the Thyroid Gland

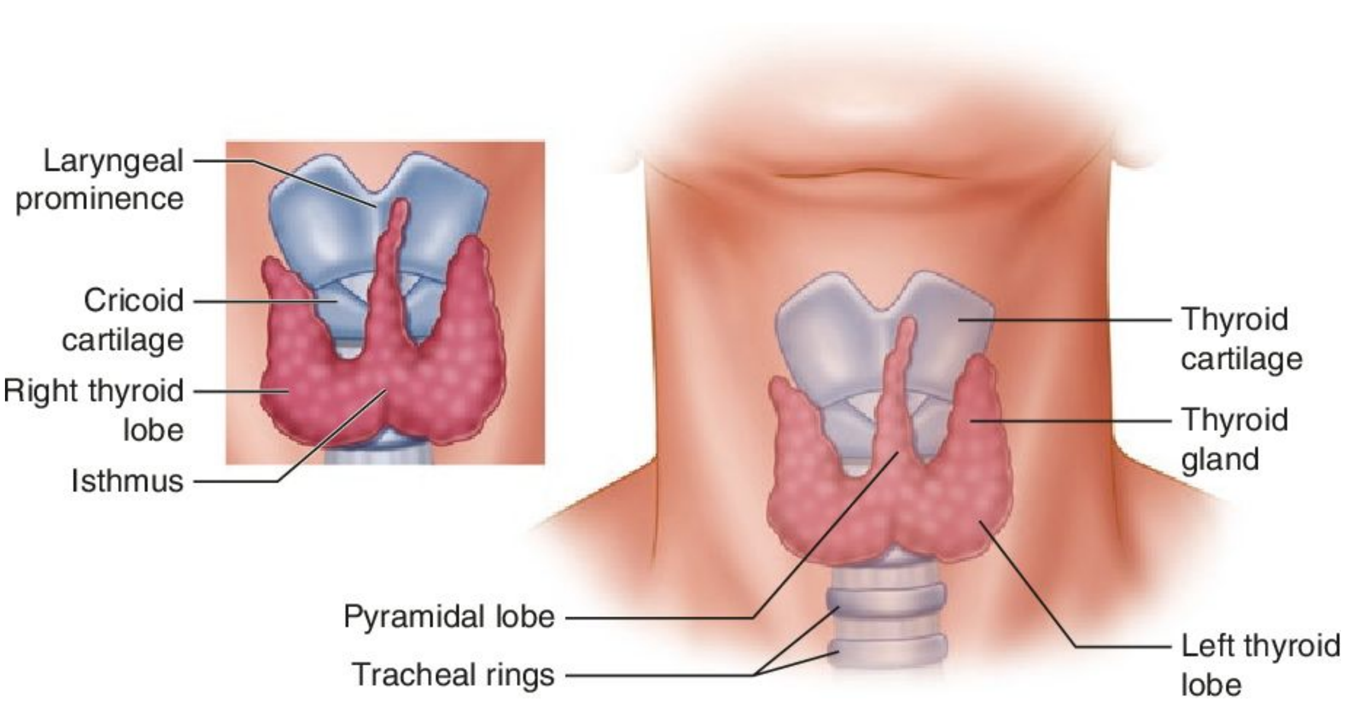


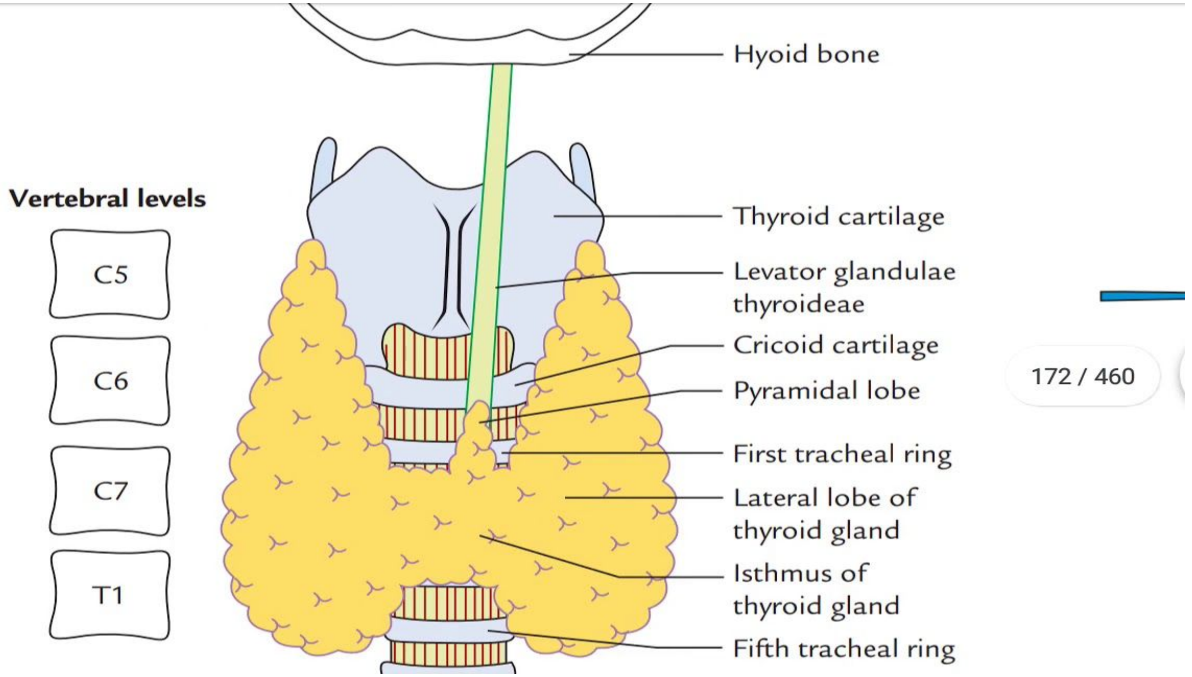
Introduction:

- The thyroid is an endocrine gland that lies in front and sides of the neck.
- The gland consists of right and left lobe joined to each other by isthmus.
- A small pyramidal lobe may project from isthmus or any of the lobes.
- Sometimes a fibromuscular band, levator of thyroid gland may project from isthmus/pyramidal lobe.

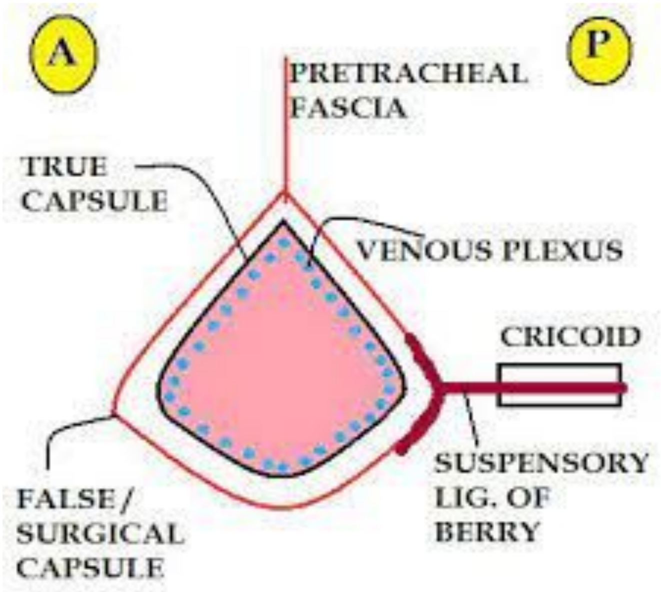


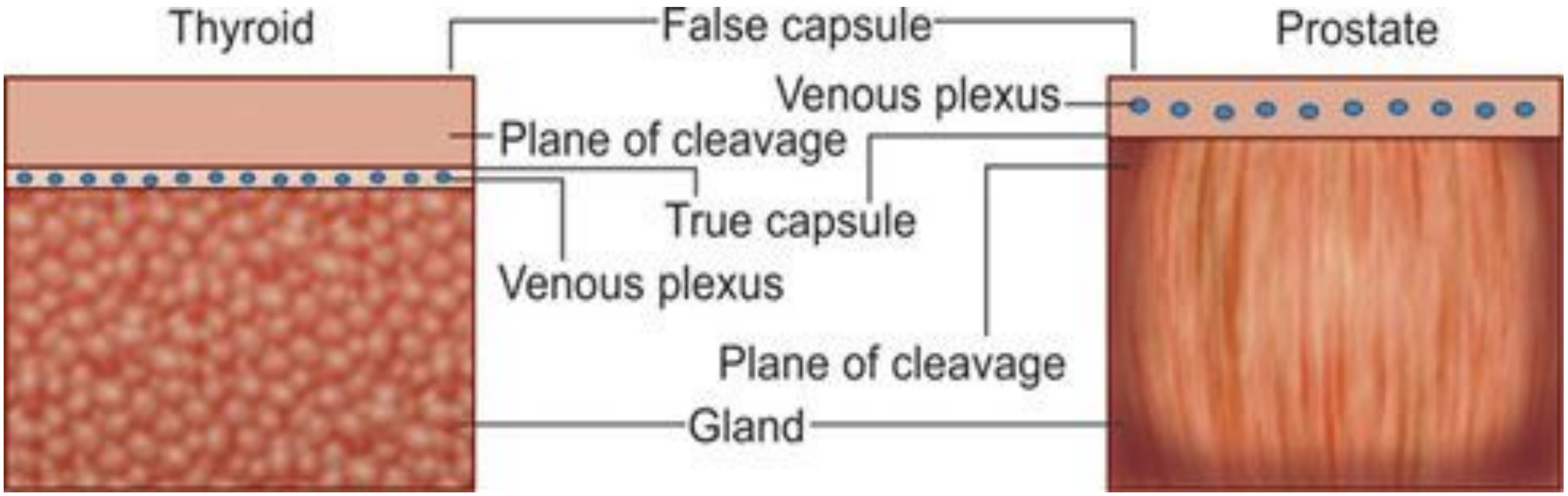
- Each lobe of the gland lies against vertebrae C5 to C7.
- The isthmus lies against 2nd to 4th tracheal ring.
- The gland is surrounded by an inner true capsule an d an outer false capsule derived from pretracheal layer of deep cervical fascia.
- The false capsule is thickened on the inner surface of thyroid gland to suspensory ligament of Berry.
- The ligament of Berry connects each lobe of thyroid gland to cricoid cartilage.



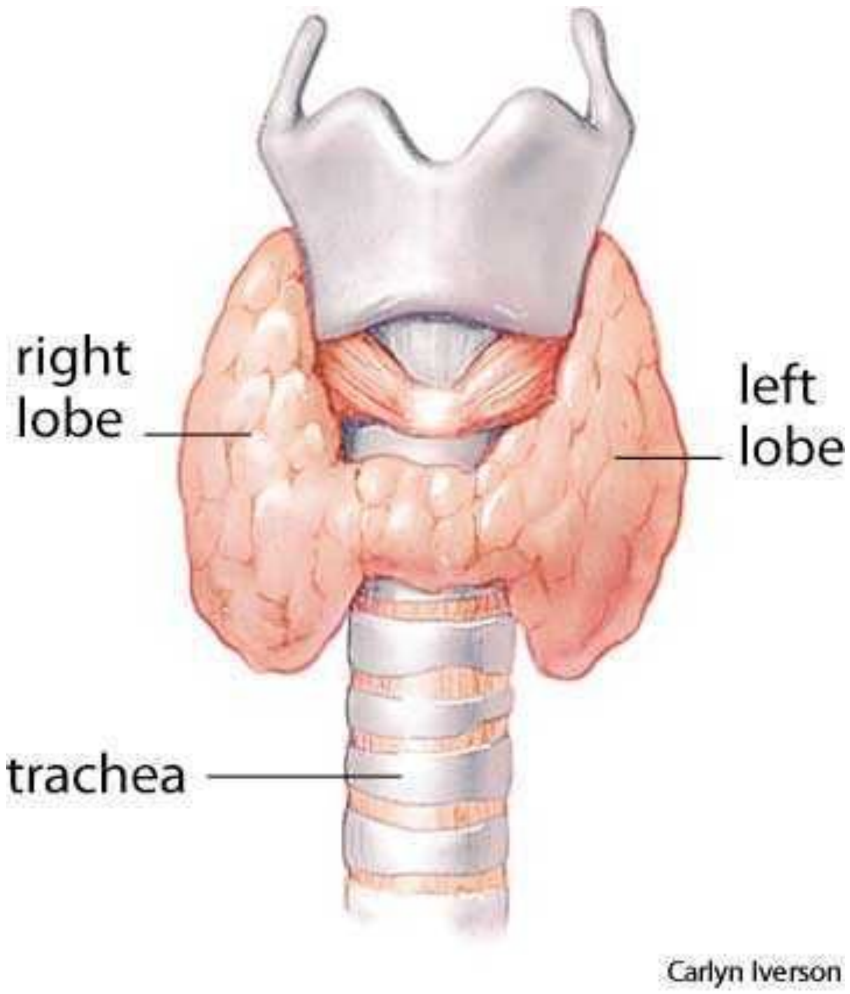


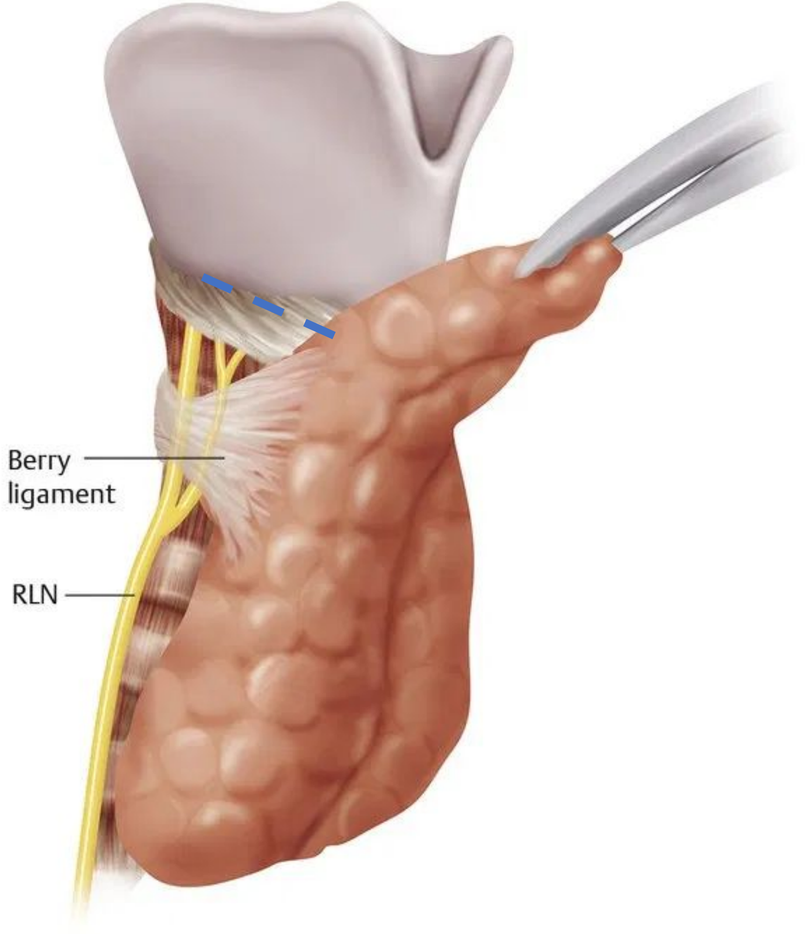
- A capillary plexus lies deep to the true capsule.
- To avoid hemorrhage during operations the thyroid is removed along with true capsule.



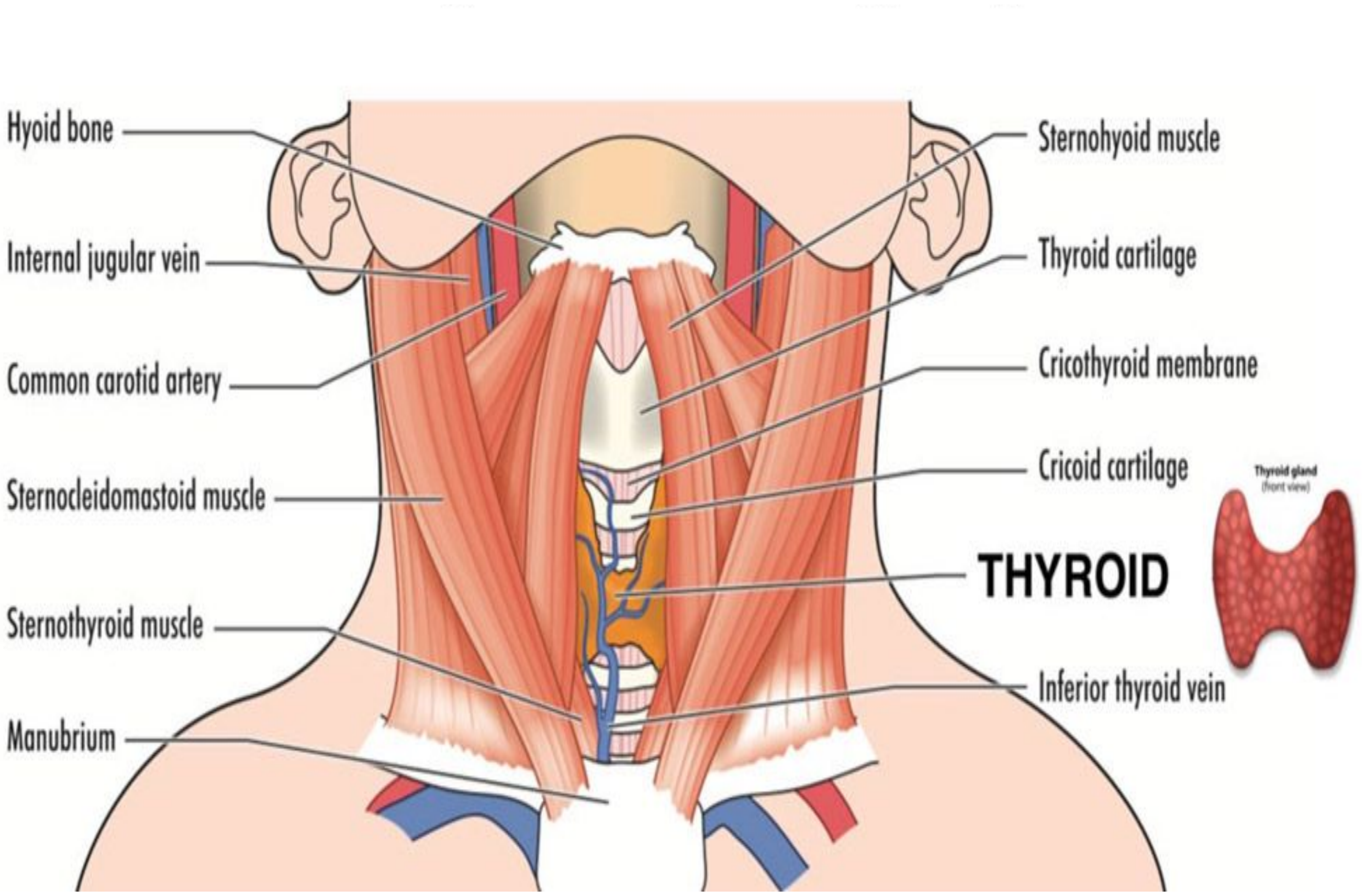


- Each lobe is conical in shape with an apex and a base, &
- Two borders: anterior and posterior, &
- Three surfaces: medial, lateral & posterior.
- Apex extends up to an oblique line on thyroid cartilage and base up to 4th or 5th tracheal ring.

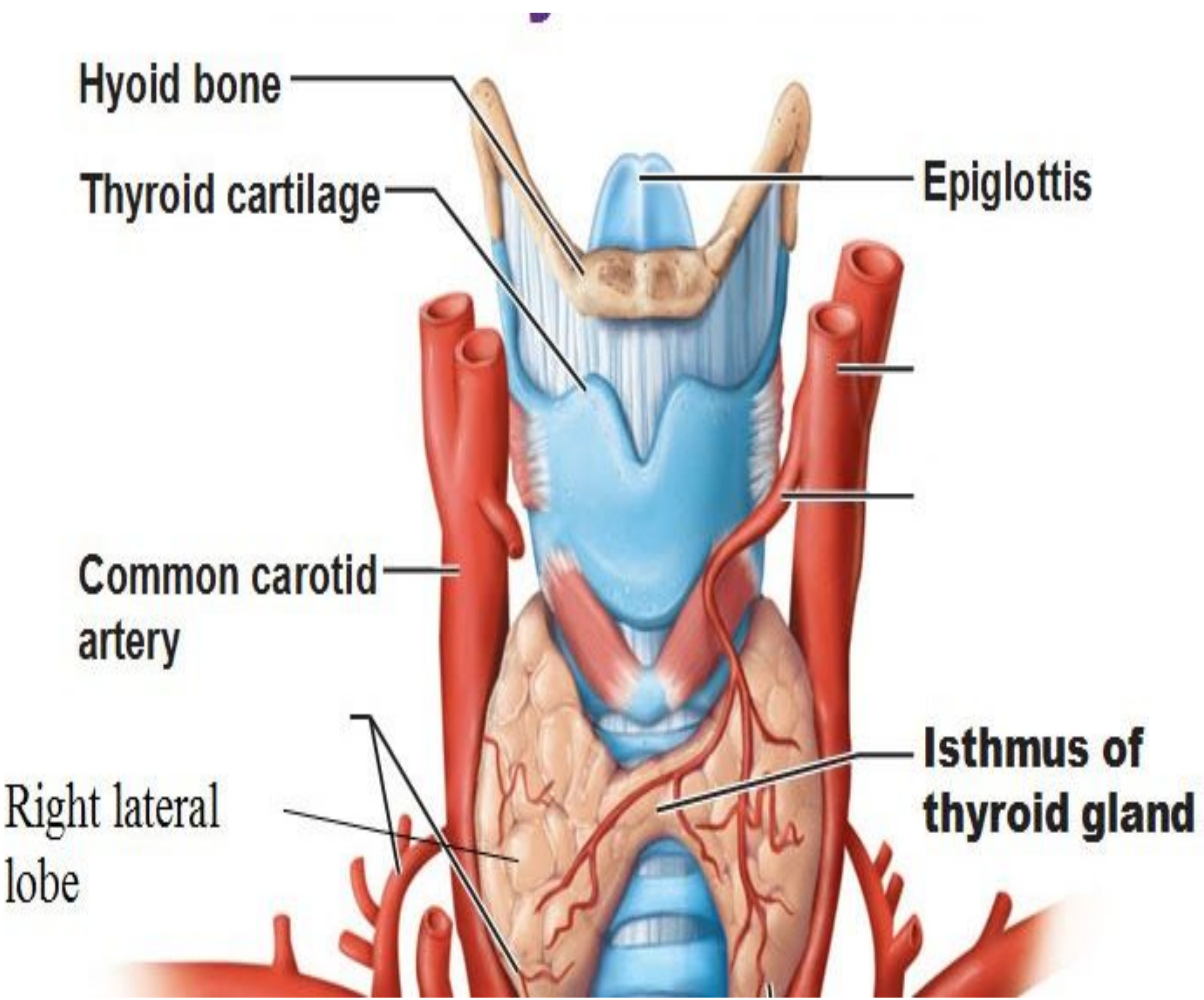




- The lateral surface is covered by infrahyoids & sternocleidomastoid.
- The medial surface is related to trachea & esophagus and external & recurrent laryngeal nerves.
- The posterior surface is related to carotid sheath.



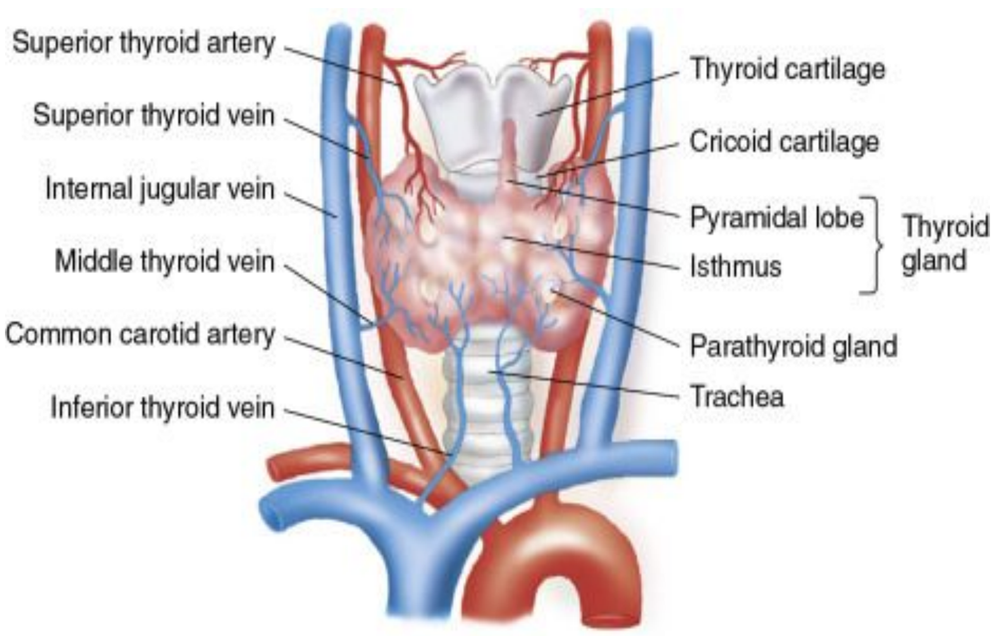
- The isthmus has upper & lower borders and anterior & posterior surfaces.
- The anterior surface is related to skin & fascia, anterior jugular veins & infrahyoids.
- The posterior surface is related to second to fourth tracheal rings.

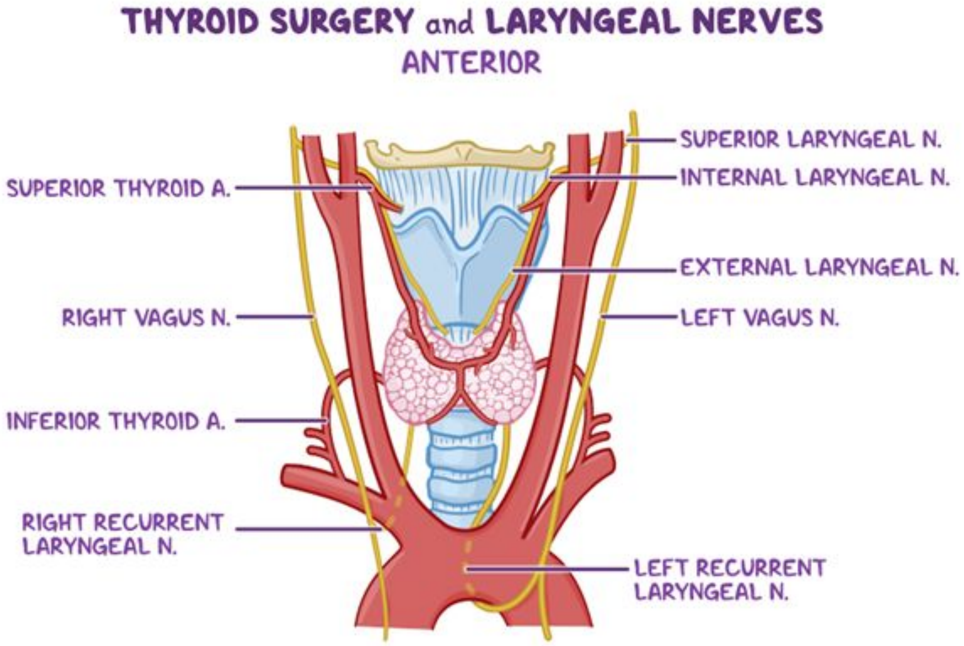


Arterial supply:

The thyroid gland is supplied by the:

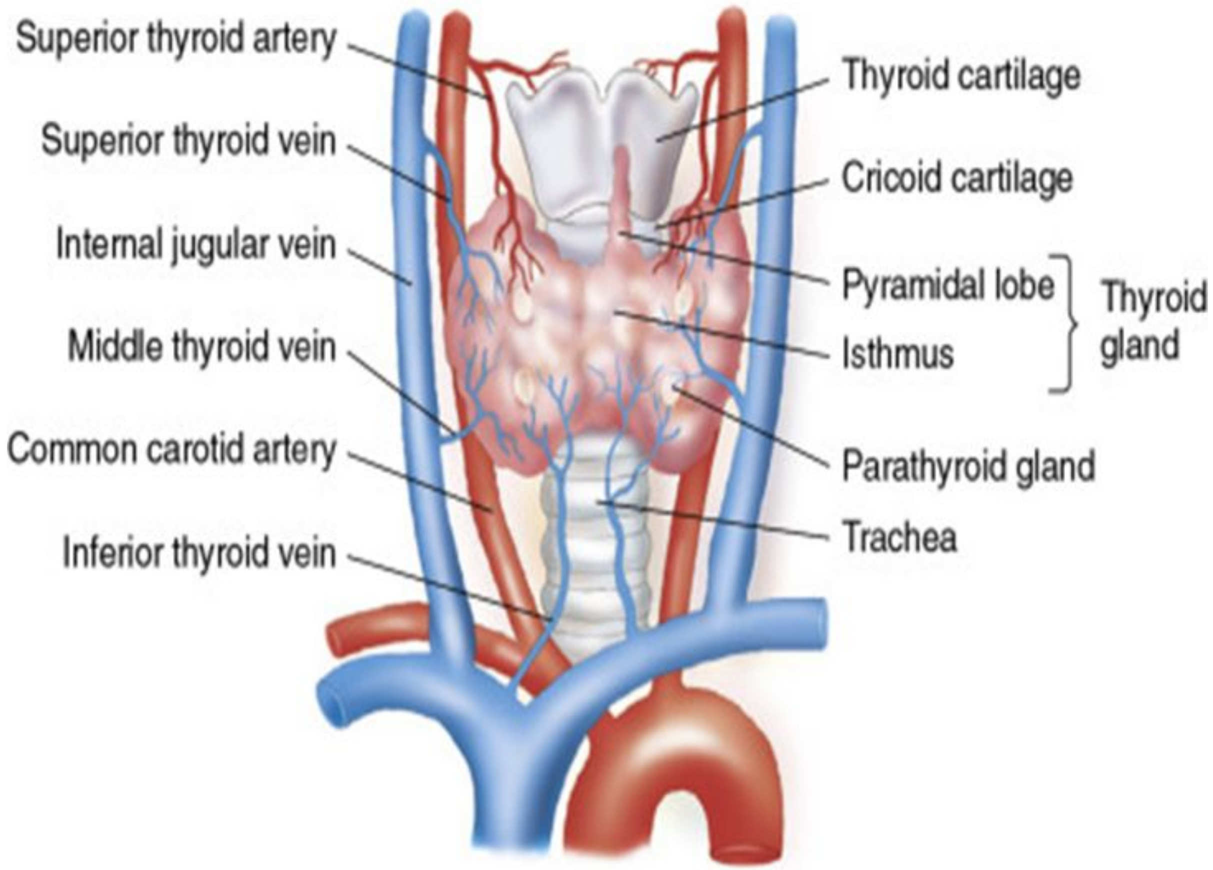
- **Superior thyroid artery:** is a 1st anterior branch of external carotid artery.
- It is in close relation to external laryngeal nerve.
- It pierces pretracheal fascia to reach apex of the lobe.
- At the upper pole, it divides into anterior & posterior branches.
- The anterior branch descends and reaches the upper border of isthmus to anastomose with its fellow on opposite side.
- The posterior branch descends along the posterior border to anastomose with the branch of inferior thyroid artery.
- It supplies upper one-third of lobes and upper half of isthmus.
- The superior thyroid artery is closely related to the external laryngeal nerve in its proximal part.
- So, during thyroidectomy, this artery should be ligated near the gland to save external laryngeal nerve.





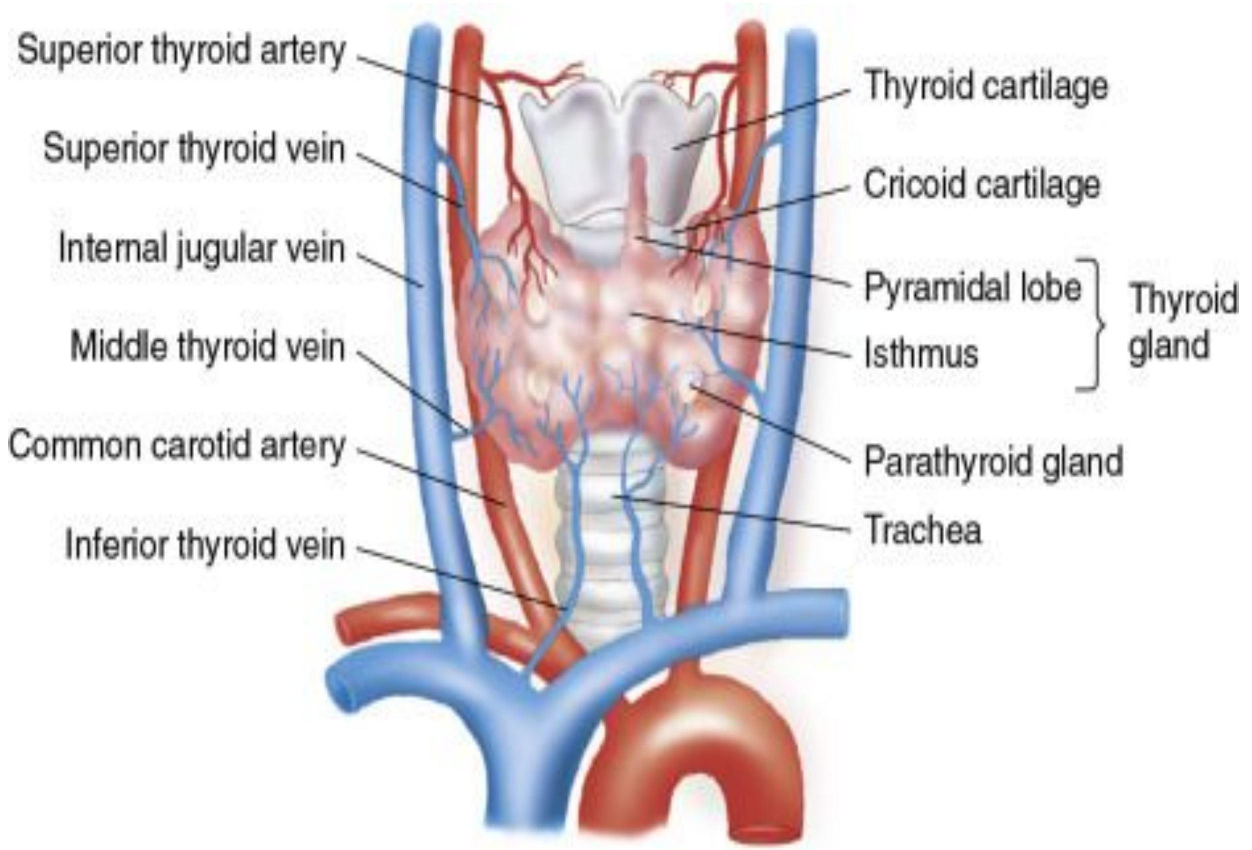
Arterial supply (continued):

- **Inferior thyroid artery** is a branch of thyrocervical trunk which arises from subclavian artery.
- It passes behind the carotid sheath to reach the base of the gland & gives 4-5 branches.
- An ascending branch anastomosis with posterior branch of superior thyroid artery.
- It supplies lower two thirds of lobes and lower half of isthmus.
- The inferior thyroid artery is intimately related to the recurrent laryngeal nerve in its terminal part.
- So, the glandular branches should be ligated away from the gland to secure the recurrent laryngeal nerve.
- Sometimes a third branch called **thyroidea ima** arises from brachiocephalic artery or arch of aorta.



Venous drainage:

- The thyroid is drained by **superior, middle and inferior thyroid veins.**
- The **superior and middle thyroid veins drain into internal jugular vein** and **inferior thyroid veins drain into right & left brachiocephalic veins.**
- A **fourth (Kocher) vein** may be present between superior & middle thyroid veins & drain into internal jugular vein.



Nerve supply:

- Mainly from middle cervical(sympathetic) ganglion, &
- Partly from superior and inferior cervical ganglia.
- These are vasomotor.

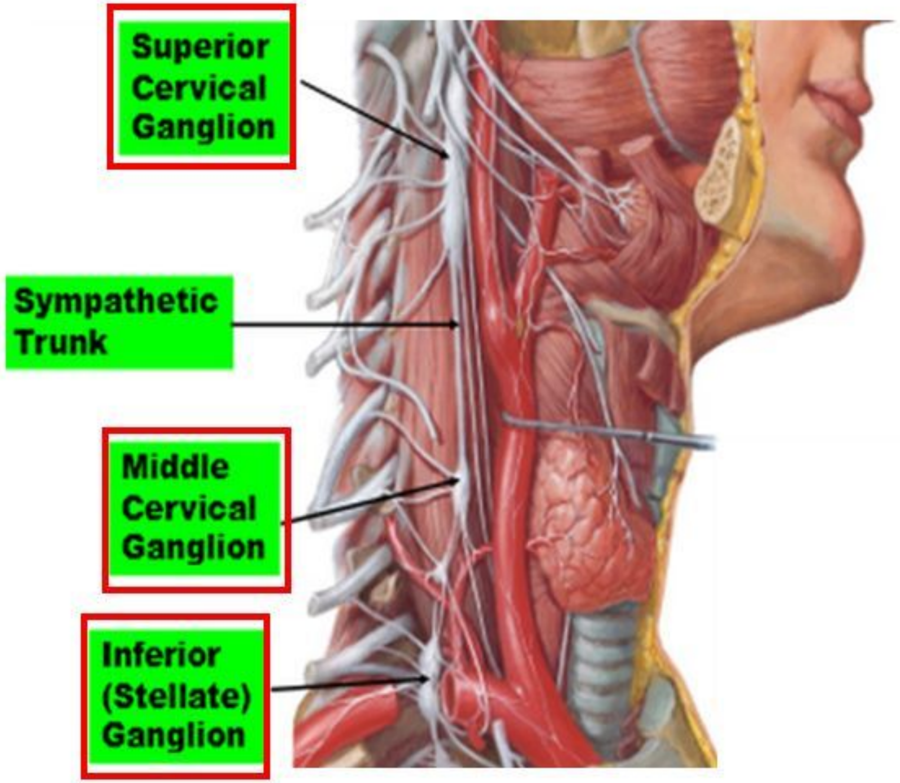
Innervation of Thyroid Gland

•Derived from the superior, middle, and inferior cervical sympathetic ganglia

•Nerves reach the thyroid through:

• Cardiac periarterial plexus
•Superior and inferior thyroid plexus

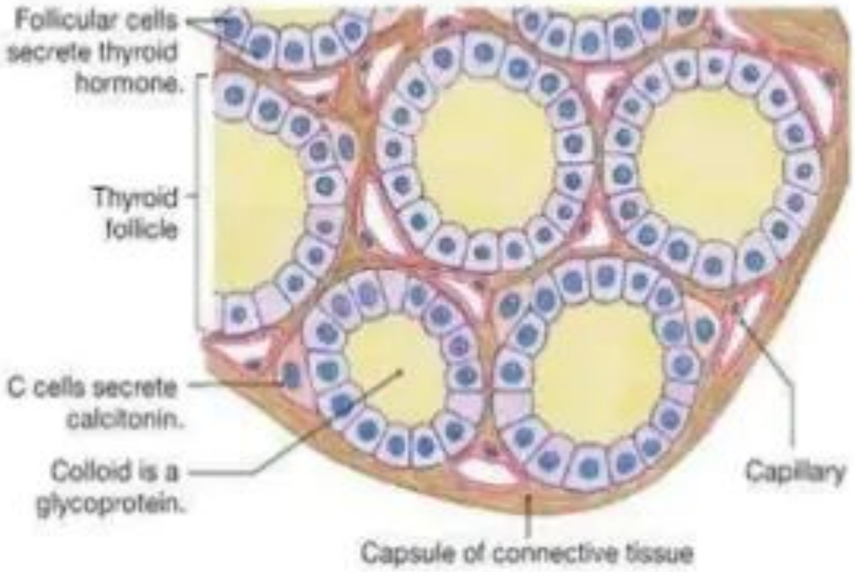
•Only vasomotor fibers causing constriction of blood vessels



•Endocrine secretion from the thyroid gland is hormonally regulated by the pituitary gland through TSH!

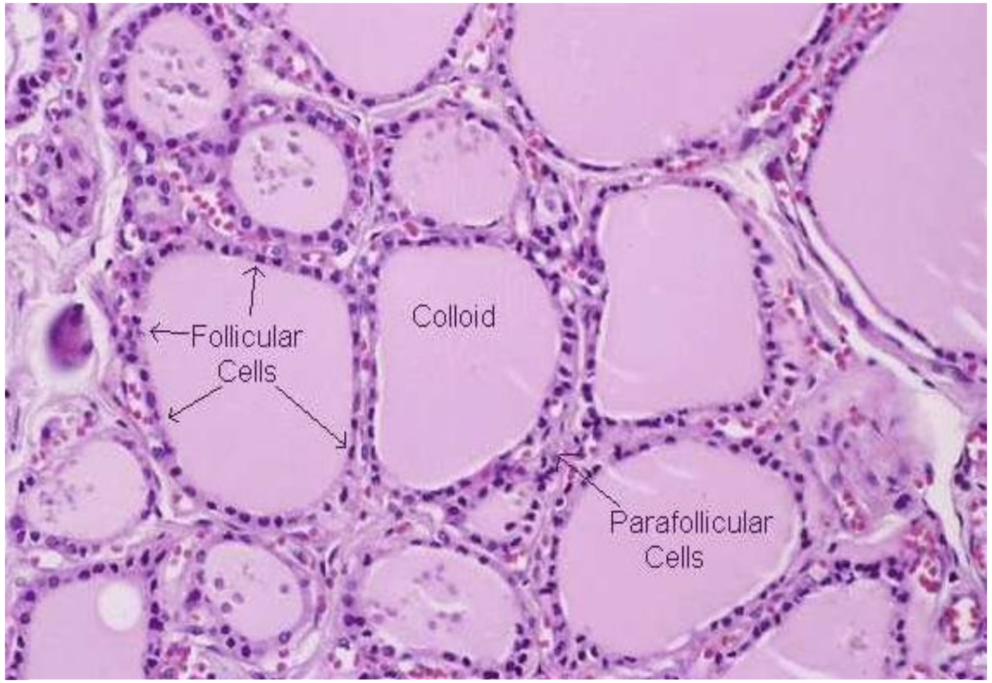
Microscopic structure

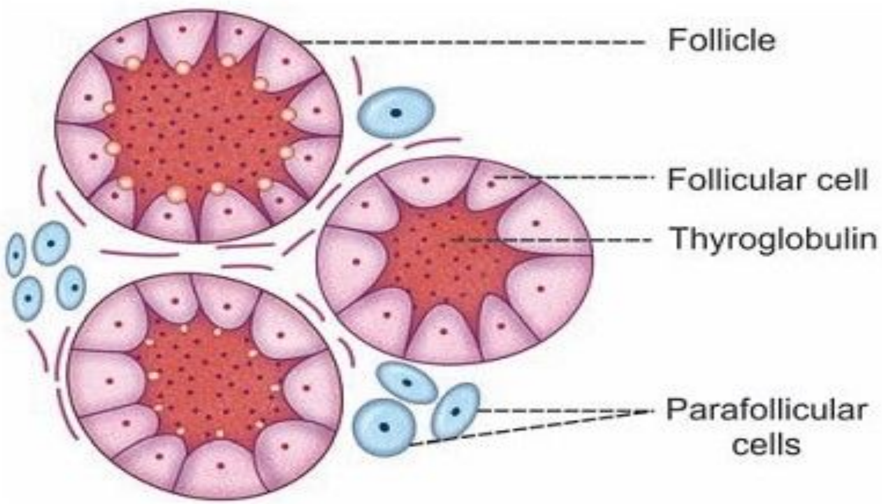
- Structural unit is follicle or acinus.
- Follicle consist of layer of simple epithelium, enclosing cavity called the follicular cavity.
- The cavity is usually filled with gel-like viscous iodine-rich material called colloid.
- Interfollicular spaces are filled by reticular connnective tissue, adipose tissue and blood vessels.



Follicular cells

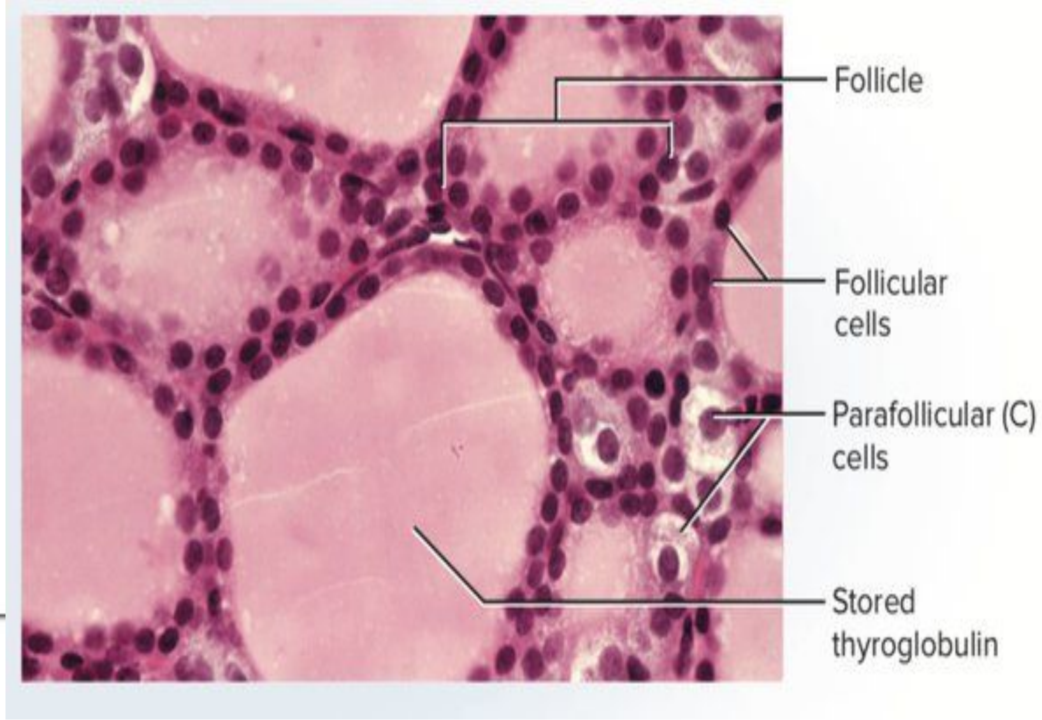
- These are cuboidal epithelial cells with their basal ends resting on basement membrane.
- These cells show the changes in shape depending on state of gland.
- When gland is inactive, cells exhibit squamous structure and columnar when hyperactive.
- The follicular cells show central or basal round nucleus with one or more excentric nuclei.
- The apical tips of cells extend microvilli in the cavity.





Parafollicular cells

- In Interfollicular spaces spaces there are some special parafollicular cells.
- They found in singly or in groups.
- They secrete hormone thyrocalcitonin, which lowers the calcium level.

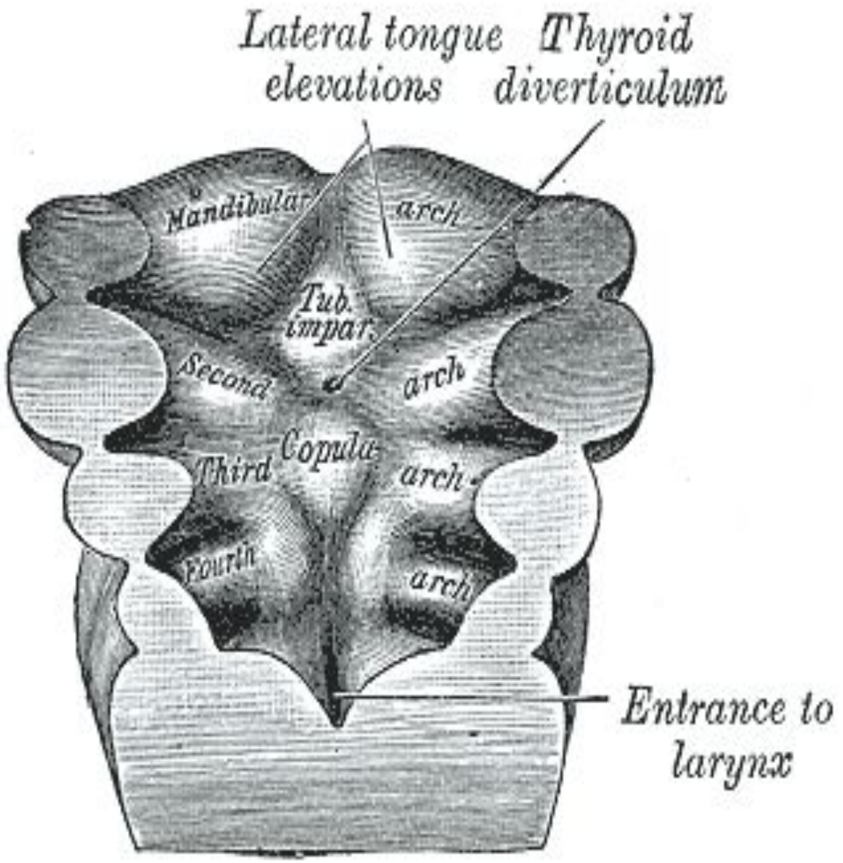


Thyroid gland malfunctioning

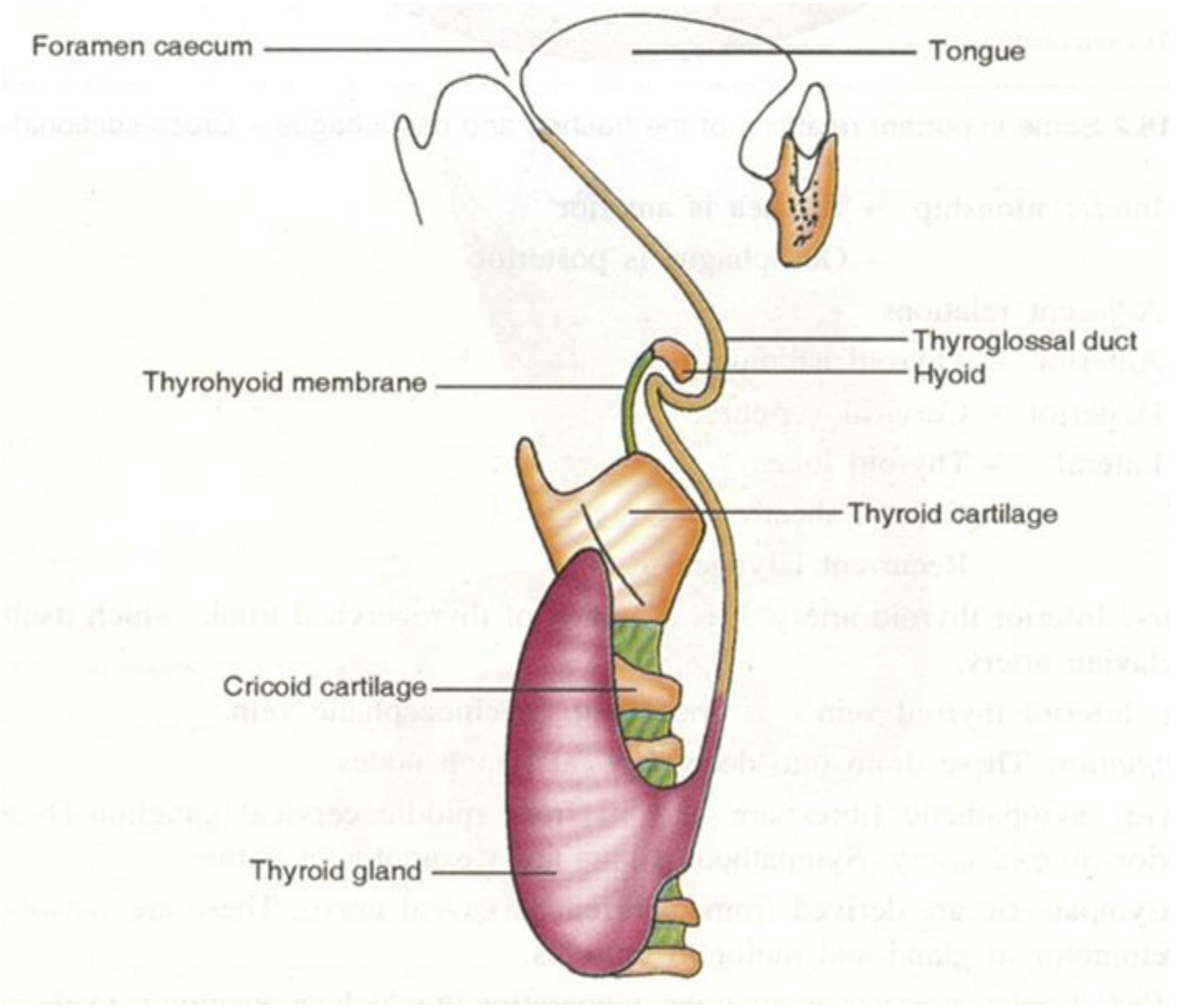
- *Hypothyroidism*
- *Cretinism.*
- *Myxedema*
- *Hyperthroidism*
- *Grave's Disease*

Development of Thyroid gland:

- It is developed as an endodermal proliferation in the floor of pharynx.
- The thyroid diverticulum appears between the tuberculum impar and copula at a point that later on remains as foramen caecum.
- The thyroid descends as a bilobed diverticulum in front of the pharynx.
- During this descend it stays connected with the tongue by a narrow thyroglossal duct.



- The thyroglossal duct afterward disappears.
- The developing thyroid gland descends ventral to the hyoid bone and laryngeal cartilages.
- Finally the bilobed thyroid with the connecting isthmus reaches in front of trachea (its final location) in 7th week.
- The colloid containing follicles appear and begin to function by the 3rd month.



- Follicular cells secrete colloid which serves as a source of T3 & T4.
- Whereas parafollicular or C-cells come from the ultimobranchial body & produce calcitonin.

Thyroglossal cyst:

- It is a cystic remnant of the thyroglossal duct.
- 50% of these may be found close to the hyoid bone.
- They may also be present at any point along their descent such as base of tongue or close thyroid cartilage.